

Deepika Pingali

Ph.D. Student, Agricultural and Applied Economics
University of Illinois Urbana-Champaign

Contact

414 Mumford Hall, University of Illinois Urbana-Champaign, Urbana, IL 61801
(415) 568-6987 • dsp7@illinois.edu
github.com/DeepikaPingali • sites.google.com/view/deepikapingali

Education

Ph.D. in Agricultural and Applied Economics Department of Agricultural and Consumer Economics (ACE), University of Illinois Urbana-Champaign GPA: 4.0/4.0	2022 – Present
Master of Business Administration Indian Institute of Management (IIM) Ahmedabad, India	2014 – 2016
B.E. (Honors), Electronics & Communication Engineering Birla Institute of Technology and Science (BITS) Pilani, India	2010 – 2014

Research Fields

Development Economics • Data Science • Political Economy • Economics of Networks

Technical Skills

Quantitative Skills: Data Science, Machine Learning, Causal Inference, Geospatial Analysis, Social Network Analysis, Network Econometrics
Software: Stata, R, Python, MATLAB

Publications

- Lyons, A., Kass-Hanna, J., **Pingali, D.**, Soliman, A., Zhu, D., Zhang, Y., & Montoya Castano, A. (2025). A geospatial analysis of food insecurity among refugee households in Lebanon using machine learning techniques. *Food Security*. (Earlier version circulated as ERF Working Paper No. 1729, Economic Research Forum.)

Working Papers

- Pingali, D.**(2026). *Uncovering the influence of hidden social networks on gendered labor norms: Evidence from rural India* [Job Market Paper].
Abstract: Economics has traditionally characterized spousal labor sharing as the outcome of an intrahousehold bargaining process; however, these decisions are also shaped by norms embedded within and transmitted through social networks. This paper investigates the influence of peer behavior on labor sharing between spouses. To do this, I jointly estimate latent social structures,

peer effects, and contextual effects via a linear social interactions model, employing an Adaptive Elastic Net Generalized Method of Moments approach to recover the structure of the network using microdata for rural Indian households. I find that the peer networks governing household production and domestic labor sharing differ in both their topology and the size of the spillover effects that they generate. Domestic labor sharing exhibits a strong social multiplier: peers' male-to-female domestic labor ratio passes through to a couple's own ratio at 0.58, while the passthrough for the female-to-male production ratio is more modest, at 0.54. More caste-diverse villages form denser production-norm networks, and densely connected sub-clusters within networks do not mirror caste or religious divisions. Women who have ever belonged to a self-help group (SHG) are central influencers in production-norm networks. These findings provide novel insights on how network-mediated spillover effects shape intrahousehold labor distributions, and more broadly, on how peer effects operate through the social networks in which households are embedded.

- **Pingali, D.**(2025). *The effect of political capital on migration decisions in the context of climate change in India* [Working paper]. Available at SSRN: <https://ssrn.com/abstract=5500739>

Abstract: Climate change is increasingly reshaping livelihoods through droughts, crop failures, rising sea levels, and extreme weather events, particularly in South Asia, Sub-Saharan Africa, and Latin America. Migration has become a crucial household strategy for managing these climate-induced risks. This paper examines how village-level political capital — households' close connections to local government officials — shapes migration decisions under environmental stress, and how this influence varies with the intensity of long-term climate stress. We develop a stylized expected-utility framework in which households weigh expected gains from migration against the costs and risks of remaining in place. The framework predicts a nonlinear, inverted U-shaped relationship: political capital has the largest effect at moderate levels of climate stress. Using data from the India Human Development Survey and ERA5 climate records, we implement a difference-in-differences approach. Empirical results confirm the model: at medium heat stress, political capital increases migration probability by 3.2 percentage points ($p < 0.05$), while its effect is smaller and statistically insignificant under low or extreme stress. These findings demonstrate that political access mediates climate adaptation, with implications for equitable migration policies and resilience strategies.

- Lyons, A., Polato e Fava, A., Kass-Hanna, J., **Pingali, D.**, & Liu, F. (2025). *Gender disparities in financial and digital financial inclusion: Insights from PISA data* [Working paper, under review]. Available at SSRN: <https://ssrn.com/abstract=5384658>

Abstract: The global digital transformation of financial services has heightened concerns about gender disparities in financial inclusion and empowerment, particularly as gender gaps in financial literacy often emerge early in life. This study leverages data from the 2018 PISA assessment to examine gender differences in the relationship between financial literacy and both traditional and digital financial inclusion among youth. Using the Lewbel method to address endogeneity, the analysis reveals that financial literacy significantly enhances financial and digital financial inclusion. However, persistent gender disparities remain, especially in traditional financial inclusion, where girls are less likely than boys to own bank accounts or payment cards. Gaps in confidence are also notable, with girls up to 50 percentage points less likely to feel confident performing financial and digital financial tasks compared to boys. Encouragingly, participation in a financial literacy course significantly improves girls' confidence, highlighting its potential to reduce these disparities. Yet greater confidence does not necessarily translate to increased financial service participation, suggesting the continued presence of external barriers to financial inclusion. These findings underscore the need for targeted policy interventions to promote

early financial education, digital skill development, and confidence-building among girls, laying a foundation for their economic empowerment in the digital age.

Works in Progress

- Lyons, A., Soliman, A., & **Pingali, D.** *Using Machine Learning Techniques to Predict Internal Migration in the Brazilian Amazon.*
- Tripathy, P., Upadhyay, S., Alegbeleye, O. M., Senkardesler, E., **Pingali, D.**, Goddard, E., Thaker, J., Leiserowitz, A., & Marlon, J. *Mismatch Between Perceived and Assessed Flood Risk in India.*
- Lyons, A., Telles, T. S., Plassa, W., Soliman, A., **Pingali, D.**, & Storoe, B. *From Forest to Farm? Deforestation, Farm Lifecycles, and Long-Run Agricultural Sustainability in Brazil.*
- Lyons, A., **Pingali, D.**, Bhatia, A., Chen, C., Mazumdar, A., Xu, Z., & Soliman, A. *Where Refugees Settle: Spatial Dependence and the Determinants of Syrian Refugee Distribution in Lebanon.*

Conference & Poster Presentations

- “The Effect of Political Capital on Migration Decisions in the Context of Climate Change in India” (sole author).
 - Political Economy of Climate and Environment (PECE), Harvard University, Cambridge, MA, September 2026 (scheduled, poster)
 - Environmental and Climate Mobilities Network (ECMN), Wageningen University & Research, Wageningen, Netherlands, June 2026
 - European Political Science Society (EPSS), Belfast, United Kingdom, June 2026
 - Illinois Economics Association Conference, Chicago, IL, October 2025
- “Gender Disparities in Financial and Digital Inclusion: Insights from PISA Data.”
 - Congreso Internacional en Alfabetización Financiera, Online, November 2025
 - International Conference on Empirical Economics, Penn State Altoona (Online), August 2025
 - Illinois Economics Association Conference, Chicago, IL, October 2024
- “A Geospatial Analysis of Food Insecurity Among Refugee Households in Lebanon Using Machine Learning Techniques.”
 - ERF 30th Annual Conference, Rabat, Morocco, April 2024 (presented by A. Lyons)
 - Illinois Economics Association Conference, Chicago, IL, October 2024

Research and Teaching Experience

Senior Research Associate

2022 – Present

Department of Agricultural and Consumer Economics (ACE) & National Center for Supercomputing Applications (NCSA), University of Illinois Urbana–Champaign

- Leads research on forced displacement, climate change, and the socioeconomic vulnerability of disadvantaged populations, producing peer-reviewed publications, working papers, and presentations at international venues.
- Mentors and manages undergraduate researchers through NCSA’s REU (Research Experiences for Undergraduates) and SPIN (Students Pushing INnovation) programs, as well as graduate

students; mentees have gone on to co-author peer-reviewed publications and working papers arising from this research.

Teaching Assistant, ACE 474: Economics of Consumption

Fall 2022

Department of Agricultural and Consumer Economics (ACE),
University of Illinois Urbana-Champaign

- Guided student teams through a Consumer Price Index (CPI) project modeled on U.S. Bureau of Labor Statistics methodology, advising them on constructing a custom price index to capture how college students' consumption baskets have shifted amid recent tariffs and rising inflation.
- Graded assignments and exams and held regular office hours to support student learning.

Teaching Assistant, ACE 270: Consumer Economics

Spring 2024

Department of Agricultural and Consumer Economics (ACE),
University of Illinois Urbana-Champaign

- Managed grading responsibilities for assignments and exams.

Fellowships & Awards

- | | |
|--|-------------|
| • Graduate Conference Travel Award, ACE | Summer 2026 |
| • Wilken Farm Management Fellowship, ACE | Spring 2025 |
| • University Block Grant Fellowship, ACE | Spring 2025 |
| • Linse Scholarship, ACE | Spring 2024 |

Training and Workshops (Competitive)

- | | |
|---|------|
| • Winter School on Econometrics of Networks, Vrije Universiteit Amsterdam, Online | 2025 |
| • Oxford Summer School on Economic Networks, Oxford, UK | 2025 |
| • I-GUIDE Summer School, Spatial AI for Extreme Events and Disaster Resilience, UCAR, Boulder, CO (NSF-supported) | 2025 |

Other Professional Experience

Over five years in the international development sector across South Asia: managing grassroots programs and a social business, and advising INGOs and multilateral agencies. Areas of work include monitoring, evaluation and learning (MEL), program design and implementation, advisory services, fundraising, and branding & marketing.

Social Entrepreneurship

Founder, Vidhignya Enterprises Pvt. Ltd.

2017 – 2022

- Founded a venture promoting collective entrepreneurship among women from the informal economy, empowering them through skill-building and livelihoods interventions and connecting them to sustainable market linkages.

Development Consulting

2017 – 2022

- World Bank: land contracts and migration trends in India.
- World Bank & Access Development Services: supply-side financing barriers for women entrepreneurs.
- International Labour Organization: worker networks and labor conditions, New Delhi.
- UNDP: vision document for May Doe Kabar, a network of 20,000 rural women in Myanmar.

- DFID Global Innovation Fund: case studies of Indian firms.
- EU GENIE Program: review of the role of women's civil society organizations in India.
- Bangladesh-based INGO: design and review of a community empowerment program.
- Evaluation of international labor migration projects across Lebanon, Jordan, and South Asia.

Program Design & Implementation

2017 – 2022

- Directed grassroots anti-trafficking projects in Nepal (UN Trust Fund-funded NGO partnership), rescuing and rehabilitating trafficking survivors and supporting informal-sector workers through community networks.
- Directed a women's empowerment project in Varanasi and Koppal, India, benefiting 7,000 rural women through self-sustaining collectives.

Private Sector Experience

True North Private Equity, Mumbai

2016 – 2017

- Drove business development initiatives, launching and expanding new business verticals and shaping corporate strategy.

References

Available upon request.